

FLOW

1. Someone open the door
2. Aqara get trigger - Send the webhook Trigger to n8n
 - a. N8n save the detail in trigger table
3. Wait 3 minutes
4. N8N search the entry method in database, then update the trigger table
 - a. If it is whitelist, no need to run LLM Input , change session to 'whitelised'
 - i. The end the flow
 - b. If it do not have any entry method
 - i. The end the flow (Need to have future to determine who responsible)
5. Retrieving Video from Entrance Da Hua CCTV
 - a. { start-time: , end-time: }
 - b. Call API to retrieve video & save Video
 - i. Insert into video_assets information
 - ii. Insert the video into Space
 - iii. Once the video is ready, update the video to 'ready'

Theft Detection System (flow for each Location)

6. Schedule Poll
7. Retrieve the video detail
 - a. Check if there is processing video or issue (If the status keeps running in processing, then how ?)
 - b. If yes then stop
 - c. Get the latest ready video
8. Run script on the video
9. Entry Script: -
 - a. Get all the active embedding into memory.
 - b. RUn the script on the video
 - c. Detect Enter
 - i. Create Session
 1. {entry_trigger_id: , location_id: , start_time: }
 - ii. Create session_customer
 1. {persion_id, enter_time}
 - iii. Save ReID Image
 1. Upload Image into Space
 - iv. Create the Embedding Value into Database - table Customer_Gallery
 - v. Insert into Active Gallery
 - d. Detect Exit
 - i. Find related Session & Customer
 - ii. Save ReID Image
 1. Upload Image into Space
 - iii. Update Customer Gallery
 - iv. Update session_customer
 1. {leave_time}
 - v. Check if all customer left
 1. If all left, Update Session

- a. Update Closed Time
 - b. Update Video Link URL for Close
- 10. Detect session is closed
- 11. Retrieve Kiosk Video between Start Time and Closed Time
- 12. Retrieve the Customer Gallery that is related to the session.
- 13. Do Kiosk Script:
 - a. Get the transactions happen between Start Time and Closed Time
 - b. Detect the related customer to do detection via embedding
 - c. Total Item Brought out from store
 - i. Update to Session
 - d. Compare the Total Item between the Kiosk and Transactions
 - e. Condition:
 - i. If Match
 - 1. Update Session set **Not Detected**
 - ii. If Kiosk quantity more than Transactions
 - 1. Update Session set **Detected**
 - iii. If Transactions quantity more than Kiosk
 - 1. Update Session set **Need Review**
- N8n will only handle Entry Video Flow (Until Step 9)
- Once the session is closed. The process will run background of the server.

** If error in on of the components, stop the rest, unless reset to the session i want.

Database:

- Whitelist_Entry
 - ID
 - Method
 - Entry_ID
 - Status
 - Created_at
 - Updated_at
- CCTV
 - ID
 - LocationId
 - Delayed Time
 - Section
- Trigger
 - ID
 - LocationId
 - Created_Time
 - Status
 - Pending
 - Done
 - Issue
- Session
 - Id
 - Entry Video Link

- Trigger_Id
- Start_time
- End_time
- Total_item_brought
- Actual_items_brought
- Status
 - Detected
 - Not Detected
 - Issue
 - Whitelisted
- Total_customer
- Created_at
- Session_Transaction
 - Id
 - Session_id
 - ReceiptNumber
 - Created_at
- Exit_Video
 - Id
 - Url
 - Session_Id
 - created_at
- Customer_Gallery
 - Id
 - Session_id
 - Embedding_Osnet
 - Embedding_Fashion
 - Image_URL
 - Created_at

PHASE 2

- If Supplier ? Own Stuff ? Worker ? Cleaner ?
 - Whitelist these people
- Sneak In? (Phase 2)
 - Consider to the group that he sneak in
 - Issue:
 - **How to know?**
- Open the door for his friend/others (Phase 2)
 - Consider to the group
 - Issue:
 - **How to know ?**
- Concurrent Session ? eg. P1 enter, loitering, P2 enter and grab and buy and leave.
P1 the only come and pay (Phase 2)
 - Maybe can use Concurrent to save time and energy
 - **How?**
- What if you open door / Aqara off the switch
 - Consider Unknown

- Product Identification (Phase 2)
 - Train Dino Grounding using product image.